HW 7 Instructions

1. First Code Drop, group assignment

- Include initial code analysis for all your data.

    - Mean, median, and mode of numerical data
    - standard deviation of one of the values - you may use Spark Scala for this
- Include code cleaning, choose at least 2 cleaning options:
    - Date formatting
    - Text formatting (removing spaces, extra characters, making all caps or all lowercase for normalization / future joining)
    - Create a binary column based on the condition of another column.
    - SPECIFY which of these **TWO** you have performed commented in your code
- An update from your Jira board, screenshot is fine. Include this assignment on your board.
- For full credit: Provide documentation of what your code does and access and details where the data is located, screenshots of the code executing.
- **Please include any updates to your team proposal that have changed.**
- **Name your file FirstCode.scala, you can use MapReduce as well, make it clear what each program does.**

<u>For full credit, provide the classes, jars, data and source files for each job and ways to access the datasource.</u>

<u>Please provide access to:</u>

pd2672_nyu_edu and adm209_nyu_edu

2. Readings from The Definitive Guide

- Spark Chapter 19

spark-shell --deploy-mode client -i FirstCode.scala

HW 8 Instructions

Your project will be deducted -15 for every day it is late. No exceptions.

1. Final Code Drop

Complete development of your code using only Big Data tools for the analytics part of your project. Project was developed on NYU's Dataproc. Turn your code in on Brightspace. Please provide:

1. A zip file containing source code for the analytic(s), class files, jar, etc. in a /ana_code directory
2. Code and commands for data ingestion in a /data_ingest directory. This can be simple or more complex.
3. ETL/cleaning code in a /etl_code directory - with subdirectories for each team member
4. Profiling code in a /profiling_code directory - with subdirectories for each team member
5. Optional: Test code and unused code in a /test_code directory
6. Screen shots that show your analytic running in a /screenshots directory - **for every step, documented. Organized in subdirectories if needed**
7. A readme file (see below) in the root (top) of your directory structure
8. Do not delete any code or data until final grades are issued.

2. Readme file - important

• Describe your directories and files, step by step, how to build your code, how to run your code, where to find results of a run.

• Identify where we can find the input data that you used. **You must provide access to the HDFS files to the same accounts from previous assignments.**

Part 1

1a

```
scala> val boroughStats = dfFiltered.groupBy("BORO_NM").agg(avg("RESPONSE_TIME_SEC").alias("avg_response_sec"), expr("percentile_approx(RESP
ONSE_TIME_SEC, 0.5)").alias("median_response_sec"), stddev("RESPONSE_TIME_SEC").alias("stddev_response_sec"), count("*").alias("num_records"
)).orderBy(desc("avg_response_sec"));
boroughStats: org.apache.spark.sql.Dataset[org.apache.spark.sql.Row] = [BORO_NM: string, avg_response_sec: double ... 3 more fields]

scala>

scala> boroughStats.show(false);
+-------------+------------------+------------------+------------------+-----------+
|BORO_NM      |avg_response_sec  |median_response_sec|stddev_response_sec|num_records|
+-------------+------------------+------------------+------------------+-----------+
|BRONX        |4553.439988301316 |1292              |7669.707089413246 |509459     |
|QUEENS       |2723.552146266656 |924               |4562.192518965853 |637083     |
|BROOKLYN     |2213.5054641974625|657               |4076.3623181104526 |867648     |
|MANHATTAN    |1756.034276880178 |591               |3200.6623023494867 |632613     |
|STATEN ISLAND|1705.3197524625339|554               |3128.6628850359675 |120709     |
+-------------+------------------+------------------+------------------+-----------+
```

```
scala> println("=== Top 10 Slowest Precincts ===");
=== Top 10 Slowest Precincts ===

scala>

scala> precinctStats.orderBy(desc("avg_response_sec")).show(10, false);
+--------+-----------+-----------------+-------------------+-------------------+-----------+
|BORO_NM |NYPD_PCT_CD|avg_response_sec |median_response_sec|stddev_response_sec|num_records|
+--------+-----------+-----------------+-------------------+-------------------+-----------+
|BRONX   |47         |6622.629356827477|1922               |10310.109110183199 |49147      |
|BRONX   |46         |6393.960706313952|1774               |10287.459449440803 |45249      |
|BRONX   |43         |5196.657627186487|1604               |8165.578590167324  |49966      |
|BRONX   |45         |4961.343869297962|1786               |7536.510283696884  |30910      |
|BRONX   |40         |4764.804283327081|956                |8469.038962123479  |58646      |
|QUEENS  |106        |4352.255574235868|1267               |7240.780944129072  |36283      |
|BRONX   |44         |4260.738339283657|1106               |7179.144635280904  |52098      |
|BRONX   |42         |4099.050837481914|1277               |6650.24014514301   |40777      |
|BROOKLYN|70         |4042.273488414193|1043               |6978.166468998191  |44192      |
|QUEENS  |102        |4016.809483421241|1588               |5823.867511383168  |37729      |
+--------+-----------+-----------------+-------------------+-------------------+-----------+
only showing top 10 rows


scala> println("=== Top 10 Fastest Precincts ==="); precinctStats.orderBy(asc("avg_response_sec")).show(10, false);
=== Top 10 Fastest Precincts ===
+---------+-----------+-----------------+-------------------+-------------------+-----------+
|BORO_NM  |NYPD_PCT_CD|avg_response_sec |median_response_sec|stddev_response_sec|num_records|
+---------+-----------+-----------------+-------------------+-------------------+-----------+
|MANHATTAN|22         |375.9785238212787|14                 |872.3657712209481  |4051       |
|QUEENS   |100        |892.2448369440228|348                |1686.227670387798  |21741      |
|BROOKLYN |76         |1015.1946169772257|293                |2019.486270167544  |19320      |
|MANHATTAN|6          |1017.1105437178545|411                |1858.3972301158415 |27220      |
|BROOKLYN |78         |1068.0623486895197|283                |2119.397038767647  |28501      |
|BROOKLYN |84         |1184.2185791312272|250                |2401.2855901845046 |52672      |
|BROOKLYN |88         |1217.9664235412474|421                |2244.754554168824  |31808      |
|MANHATTAN|7          |1274.5242285237698|378                |2698.0724255012233 |23980      |
|BROOKLYN |81         |1295.4380209312528|442                |2397.958530832541  |31914      |
|BROOKLYN |79         |1303.2712906057945|406                |2470.9705284000183 |44421      |
+---------+-----------+-----------------+-------------------+-------------------+-----------+
only showing top 10 rows
```

```
=== Dispatch Delay by Borough ===
+-------------+------------------+-------------------+-------------------+-----------+
|BORO_NM      |avg_dispatch_sec  |median_dispatch_sec|stddev_dispatch_sec|num_records|
+-------------+------------------+-------------------+-------------------+-----------+
|BRONX        |1614.747389650023 |283                |3517.917032049219  |844427     |
|QUEENS       |508.48010617882943|111                |1306.7196656822457 |883792     |
|BROOKLYN     |409.26814728498186|77                 |1086.859688505097  |1259789    |
|MANHATTAN    |408.91302491088766|101                |946.0133828865453  |985834     |
|STATEN ISLAND|320.20846218956945|76                 |729.8116400418409  |180757     |
+-------------+------------------+-------------------+-------------------+-----------+
```

```
=== Top 10 Slowest Precincts by Dispatch Delay ===
+-------+-----------+-----------------+------------------+-------------------+-----------+
|BORO_NM|NYPD_PCT_CD|avg_dispatch_sec  |median_dispatch_sec|stddev_dispatch_sec|num_records|
+-------+-----------+-----------------+------------------+-------------------+-----------+
|BRONX  |46         |2707.1604920422647|665                |4963.099410858232  |83001      |
|BRONX  |47         |2445.701677670477 |514                |4824.74848064446   |81184      |
|BRONX  |44         |1986.4923389801634|380                |3879.160543906747  |93917      |
|BRONX  |40         |1893.6953938466968|256                |4064.51420592942   |101994     |
|BRONX  |43         |1637.7752461571604|317                |3362.694687336535  |78913      |
|BRONX  |45         |1447.430248354914 |258                |3294.5959459798582 |47110      |
|BRONX  |42         |1277.2352383415223|262                |2605.409103979232  |69606      |
|BRONX  |49         |1272.9210252796618|254                |2719.9309337142768 |56765      |
|QUEENS |106        |1091.3351517781498|157                |2906.4877941516934 |48393      |
|BRONX  |52         |1035.7507772395343|190                |2237.483756405942  |82986      |
+-------+-----------+-----------------+------------------+-------------------+-----------+
only showing top 10 rows

=== Top 10 Fastest Precincts by Dispatch Delay ===
+-------------+-----------+-----------------+------------------+-------------------+-----------+
|BORO_NM      |NYPD_PCT_CD|avg_dispatch_sec  |median_dispatch_sec|stddev_dispatch_sec|num_records|
+-------------+-----------+-----------------+------------------+-------------------+-----------+
|MANHATTAN    |22         |131.203219736567  |7                  |472.15900719810156 |4783       |
|QUEENS       |100        |170.24352625411106|41                 |415.08503567852364 |26453      |
|BROOKLYN     |76         |190.68069154399177|26                 |505.6442035236049  |29210      |
|BROOKLYN     |84         |196.80860854719094|23                 |559.1024134506679  |74972      |
|BROOKLYN     |78         |214.76737343545676|41                 |522.1720321371203  |42824      |
|BROOKLYN     |88         |224.0044993572347 |39                 |553.6858109043003  |42006      |
|BROOKLYN     |81         |229.48112680164306|53                 |531.2249453089263  |41143      |
|STATEN ISLAND|123        |232.4093492586491 |67                 |500.7012684465337  |24280      |
|QUEENS       |101        |237.78916777988704|70                 |491.5519890216687  |42669      |
|BROOKLYN     |79         |244.45199993434767|54                 |584.6372779463326  |60927      |
+-------------+-----------+-----------------+------------------+-------------------+-----------+
only showing top 10 rows
```

```
=== Travel Time by Borough ===
+-------------+-----------------+----------------+-----------------+-----------+
|BORO_NM      |avg_travel_sec    |median_travel_sec|stddev_travel_sec |num_records|
+-------------+-----------------+----------------+-----------------+-----------+
|BRONX        |3709.024962028083 |994              |6527.658492327748 |457575     |
|QUEENS       |2496.773420539411 |793              |4280.171425346876 |579929     |
|BROOKLYN     |2187.6215678596814|675              |3955.0196202992597|732961     |
|MANHATTAN    |1649.1292953654036|524              |3107.060622748004 |554784     |
|STATEN ISLAND|1584.324221583553 |473              |3022.502327621235 |108907     |
+-------------+-----------------+----------------+-----------------+-----------+
```

```
=== Top 10 Slowest Precincts by Travel Time ===
+--------+-----------+------------------+----------------+------------------+-----------+
|BORO_NM |NYPD_PCT_CD|avg_travel_sec    |median_travel_sec|stddev_travel_sec |num_records|
+--------+-----------+------------------+----------------+------------------+-----------+
|BRONX   |47         |5012.027570462727 |1163            |8552.11420805797  |45556      |
|BRONX   |46         |4807.470702836005 |1193            |8413.438779341714 |40550      |
|BRONX   |43         |4174.514457831326 |1113            |7032.792748070211 |46480      |
|BRONX   |45         |4005.497929463731 |1290            |6430.5902062198375|28978      |
|BRONX   |40         |3954.4200307729684|870             |7201.374354230745 |49394      |
|QUEENS  |106        |3889.721740751095 |1138            |6479.16134024629  |32193      |
|BROOKLYN|70         |3882.404420176807 |1013            |6661.125414036607 |38460      |
|QUEENS  |102        |3579.2281597868027|1281            |5415.242456880102 |36023      |
|BRONX   |42         |3463.774717857336 |988             |5997.005840978788 |37038      |
|BRONX   |44         |3421.9509836612206|913             |5999.977749419514 |44985      |
+--------+-----------+------------------+----------------+------------------+-----------+
only showing top 10 rows

=== Top 10 Fastest Precincts by Travel Time ===
+-------------+-----------+------------------+----------------+------------------+-----------+
|BORO_NM      |NYPD_PCT_CD|avg_travel_sec    |median_travel_sec|stddev_travel_sec |num_records|
+-------------+-----------+------------------+----------------+------------------+-----------+
|MANHATTAN    |22         |297.8306849315068 |9               |760.5473990631904 |3650       |
|QUEENS       |100        |858.8685033426784 |329             |1652.1817810841048|18548      |
|MANHATTAN    |6          |948.0957383716004 |363             |1799.6131089967403|23606      |
|BROOKLYN     |76         |1035.5469010650584|347             |2009.24077952835  |15586      |
|MANHATTAN    |7          |1162.519959745052 |304             |2645.005563086146 |20867      |
|BROOKLYN     |78         |1206.535493386621 |411             |2189.830666557431 |21018      |
|BROOKLYN     |88         |1218.9901636824713|451             |2206.8506812550095|26026      |
|STATEN ISLAND|123        |1223.322719333602 |510             |2094.2034018822633|14886      |
|MANHATTAN    |17         |1247.4608234457871|454             |2254.565578519392 |18289      |
|MANHATTAN    |28         |1266.5466279972222|417             |2597.9502365790563|24481      |
+-------------+-----------+------------------+----------------+------------------+-----------+
only showing top 10 rows
```

Part 2

```
=== Incident Type Impact (CIP_JOBS) ===
+------------+-----------------+-----------------+-----------------+-----------------+-----------------+-----------------+-----------+
|CIP_JOBS    |avg_response_sec |median_response_sec|avg_dispatch_sec |median_dispatch_sec|avg_travel_sec   |median_travel_sec|num_records|
+------------+-----------------+-----------------+-----------------+-----------------+-----------------+-----------------+-----------+
|CRITICAL    |489.27911168793713|257              |150.73217968130362|58               |338.5469320066335|183              |69345      |
|SERIOUS     |684.692080343489 |363              |190.40594488883661|73               |494.28613545465237|262              |136016     |
|NON CRITICAL|1592.546513644096 |657              |504.95346058137955|117              |1087.5930530627165|436              |213389     |
|NON CIP     |2905.918459171257 |913              |559.2944534184392 |92               |2347.2861418057682|601              |2348762    |
+------------+-----------------+-----------------+-----------------+-----------------+-----------------+-----------------+-----------+
```

Part 3a

```
=== Response Time by Hour ===
+-----------+------------------+-------------------+-------------------+-----------+
|hour_of_day|avg_response_sec  |median_response_sec|stddev_response_sec|num_records|
+-----------+------------------+-------------------+-------------------+-----------+
|0          |1610.3749634737887|420                |3409.273333064512  |119777     |
|1          |1413.3393896282246|365                |3449.5785148568684 |101217     |
|2          |1299.5350972182116|388                |2830.7882148927265 |83112      |
|3          |1333.6589106667423|412                |2994.991113311685  |70612      |
|4          |1359.4493516577375|441                |2989.582058611005  |64395      |
|5          |1603.9978627126852|544                |3319.625598584329  |59889      |
|6          |2042.7305472932778|631                |3903.664067378239  |67240      |
|7          |2746.790659170498 |1434               |4213.425449589309  |83954      |
|8          |2127.6615257104772|737                |4214.52248266404   |117034     |
|9          |2433.4853589561785|798                |4705.606519398269  |122703     |
|10         |2722.6243932925345|889                |5161.104724607882  |123206     |
|11         |2905.12315426668  |919                |5443.59214811623   |129691     |
|12         |3090.194303449119 |979                |5645.003116633966  |130816     |
|13         |3056.1178311199906|877                |5709.487395531701  |136144     |
|14         |3425.8325764371025|883                |5929.807586021148  |141587     |
|15         |3823.7199372865703|1740               |5895.978054170483  |139683     |
|16         |3056.6274314677707|847                |5658.4481947118    |150732     |
|17         |3151.6211412405987|885                |5640.70928515056   |144930     |
|18         |3021.121900812354 |866                |5387.965779986844  |146611     |
|19         |2858.720007609063 |803                |5229.089278316639  |141936     |
|20         |2770.846766095688 |766                |5082.924214693261  |135440     |
|21         |2724.509563402991 |755                |4942.197037552117  |126524     |
|22         |2632.7316774097335|670                |4660.562607471021  |120698     |
|23         |2731.1072631204315|1148               |4188.488995242909  |109581     |
+-----------+------------------+-------------------+-------------------+-----------+

=== Top 5 Slowest Hours ===
+-----------+------------------+-------------------+-------------------+-----------+
|hour_of_day|avg_response_sec  |median_response_sec|stddev_response_sec|num_records|
+-----------+------------------+-------------------+-------------------+-----------+
|15         |3823.7199372865703|1740               |5895.978054170484  |139683     |
|14         |3425.8325764371025|883                |5929.807586021149  |141587     |
|17         |3151.6211412405987|885                |5640.70928515056   |144930     |
|12         |3090.194303449119 |979                |5645.003116633965  |130816     |
|16         |3056.6274314677707|847                |5658.4481947118    |150732     |
+-----------+------------------+-------------------+-------------------+-----------+
only showing top 5 rows
```

```
=== Response Time by Month ===
+-----+------------------+-------------------+-------------------+-----------+
|month|avg_response_sec  |median_response_sec|stddev_response_sec|num_records|
+-----+------------------+-------------------+-------------------+-----------+
|1    |2252.193683006492 |696                |4059.0403154467067 |223809     |
|2    |2241.1977772770206|694                |4028.170727106833  |208573     |
|3    |2351.041024759861 |720                |4284.251384158808  |238612     |
|4    |2456.561534781278 |723                |4570.675742680442  |232007     |
|5    |2668.161681869686 |762                |5020.4800239792685 |245346     |
|6    |2901.0852259856197|808                |5416.208573204962  |238378     |
|7    |3141.633130298649 |852                |6073.076072066533  |233786     |
|8    |2975.5586419351903|799                |5744.596589821029  |230799     |
|9    |2900.0649524879022|796                |5404.745889772814  |233309     |
|10   |2575.918452258002 |758                |4758.227574802749  |237223     |
|11   |2518.7555927689273|756                |4728.927454183871  |224531     |
|12   |2573.674042118306 |797                |4657.687211939077  |221139     |
+-----+------------------+-------------------+-------------------+-----------+

=== Top 5 Slowest Months ===
+-----+------------------+-------------------+-------------------+-----------+
|month|avg_response_sec  |median_response_sec|stddev_response_sec|num_records|
+-----+------------------+-------------------+-------------------+-----------+
|7    |3141.633130298649 |852                |6073.076072066533  |233786     |
|8    |2975.5586419351903|799                |5744.596589821029  |230799     |
|6    |2901.0852259856197|808                |5416.208573204961  |238378     |
|9    |2900.0649524879022|796                |5404.745889772815  |233309     |
|5    |2668.161681869686 |762                |5020.4800239792685 |245346     |
+-----+------------------+-------------------+-------------------+-----------+
only showing top 5 rows

dfMonth: org.apache.spark.sql.DataFrame = [CAD_EVNT_ID: string, CREATE_DATE: date ... 19 more field
monthlyStats: org.apache.spark.sql.Dataset[org.apache.spark.sql.Row] = [month: int, avg_response_se

scala> val dfDay = dfTime.withColumn("day_of_week", date_format(col("ADD_TS"), "E")); println("=== 
).agg(avg("RESPONSE_TIME_SEC").alias("avg_response_sec"), expr("percentile_approx(RESPONSE_TIME_SEC
as("stddev_response_sec"), count("*").alias("num_records")); dayStats.show(false);
=== Response Time by Day ===
+-----------+------------------+-------------------+-------------------+-----------+
|day_of_week|avg_response_sec  |median_response_sec|stddev_response_sec|num_records|
+-----------+------------------+-------------------+-------------------+-----------+
|Thu        |2751.0324657020574|802                |5077.183378186318  |386685     |
|Wed        |2680.3014465445053|767                |5027.429013014361  |399089     |
|Fri        |2784.6350696201903|810                |5134.243462197138  |403044     |
|Mon        |2763.444455696459 |808                |5114.46735278952   |394991     |
|Sun        |2362.130340128433 |682                |4567.951023409271  |392734     |
|Sat        |2420.336893335816 |709                |4632.175746489073  |397378     |
|Tue        |2681.4520860487155|770                |5073.904288911284  |393591     |
+-----------+------------------+-------------------+-------------------+-----------+
```

3 additional

=== Hour x Borough ===

hour_of_day	BORO_NM	avg_response_sec	num_records
0	BRONX	2895.1459299681674	21990
0	QUEENS	1662.8340258630733	25751
0	BROOKLYN	1335.4419829024187	38368
0	MANHATTAN	1056.9649262516446	28882
0	STATEN ISLAND	968.7599247806102	4786
1	BRONX	2575.7887263798443	18734
1	QUEENS	1389.6541322688922	22046
1	BROOKLYN	1160.6453320802004	31920
1	MANHATTAN	966.8092635212888	24332
1	STATEN ISLAND	857.9761051373955	4185
2	BRONX	2300.7720717781403	15325
2	QUEENS	1302.9404873219296	18181
2	BROOKLYN	1068.5118592494343	25634
2	MANHATTAN	935.1149056236621	20556
2	STATEN ISLAND	716.163056206089	3416
3	BRONX	2406.877945937806	13281
3	QUEENS	1333.5699316056973	15937
3	BROOKLYN	1071.8417487684728	21112
3	MANHATTAN	951.0533115099082	17107
3	STATEN ISLAND	647.2696062992126	3175
4	BRONX	2380.784208022832	12614
4	QUEENS	1377.313192682601	14978
4	BROOKLYN	1089.787373982425	18549
4	MANHATTAN	959.9816701916299	15603
4	STATEN ISLAND	636.7680120709166	2651
5	BRONX	2985.316489135066	11735
5	QUEENS	1612.417521647384	13743
5	BROOKLYN	1247.0218613366646	17611
5	MANHATTAN	1047.7670573461698	14334
5	STATEN ISLAND	766.2976480129764	2466
6	BRONX	4217.751793843067	12961
6	QUEENS	1943.4545277885736	15438
6	BROOKLYN	1389.6514545364746	20247
6	MANHATTAN	1353.0880395512454	15777
6	STATEN ISLAND	1135.9343272985445	2817
7	BRONX	4330.39829059829	16965
7	QUEENS	2806.3140338214066	19396
7	BROOKLYN	2302.9311898169194	24197
7	MANHATTAN	2018.8177576977853	20363
7	STATEN ISLAND	1936.8206396307287	3033
8	BRONX	3912.6649250340756	22010
8	QUEENS	2157.456150920661	26557
8	BROOKLYN	1660.2669164585068	36030
8	MANHATTAN	1425.1453019050693	27873
8	STATEN ISLAND	1326.233347940403	4564
9	BRONX	4311.408654258513	23318
9	QUEENS	2479.0789293562448	27759
9	BROOKLYN	1966.5095416301722	37677
9	MANHATTAN	1660.019375109094	28645
9	STATEN ISLAND	1433.3220211161388	5304

=== Hour x CIP_JOBS ===

hour_of_day	CIP_JOBS	avg_response_sec	num_records
0	NON CIP	1716.061102572282	103449
0	NON CRITICAL	1400.7407036434206	7987
0	SERIOUS	561.9338092147956	4623
0	CRITICAL	423.7603550295858	3718
1	NON CIP	1509.4124501123904	87196
1	NON CRITICAL	1247.3379099923723	6555
1	SERIOUS	499.3945341614907	4025
1	CRITICAL	364.10403952339436	3441
2	NON CIP	1400.4036410886545	70803
2	NON CRITICAL	1068.9058588790133	5513
2	SERIOUS	491.63867344209683	3739
2	CRITICAL	367.37553156689563	3057
3	NON CIP	1444.8722071483603	59706
3	NON CRITICAL	1109.9850971922247	4630
3	SERIOUS	481.93836405529953	3472
3	CRITICAL	389.537089871612	2804
4	NON CIP	1481.8577931771688	54259
4	NON CRITICAL	1070.5676948434188	4247
4	SERIOUS	477.6558841651263	3246
4	CRITICAL	393.6613696556943	2643
5	NON CIP	1737.9441759869062	51322
5	NON CRITICAL	1193.0384615384614	3978
5	SERIOUS	524.3791650708541	2611
5	CRITICAL	380.17997977755306	1978
6	NON CIP	2192.7356401736065	58984
6	NON CRITICAL	1326.7581534221406	4354
6	SERIOUS	630.110408765276	2373
6	CRITICAL	487.1981687377371	1529
7	NON CIP	2990.4557962167873	72901
7	NON CRITICAL	1521.290053285968	5630
7	SERIOUS	793.9260106788711	3933
7	CRITICAL	610.38322147651	1490
8	NON CIP	2305.242501384584	102919
8	NON CRITICAL	1093.739878274676	7558
8	SERIOUS	544.7785501489573	5035
8	CRITICAL	490.17082785808145	1522
9	NON CIP	2652.8595133052895	107213
9	NON CRITICAL	1202.3892483237266	8501
9	SERIOUS	597.6161616161617	5346
9	CRITICAL	461.69202678028	1643
10	NON CIP	2998.0580671881326	106239
10	NON CRITICAL	1308.1054358525648	9143
10	SERIOUS	672.4447471479652	5873
10	CRITICAL	524.7022039979498	1951
11	NON CIP	3198.7978179797037	111548
11	NON CRITICAL	1477.7478377265238	9712
11	SERIOUS	694.1534956794973	6365
11	CRITICAL	570.5004840271055	2066
12	NON CIP	3424.369250372391	111442
12	NON CRITICAL	1580.8060050692143	10258